Characteristics of Macropore Transport Studied with the ARS Root Zone Water Quality Model

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ABSTRACT. The ARS Root Zone Water Quality Model components dealing with preferential water and chemical transport are presented and used to study macropore flow and transport in a silty clay loam soil. Macroporosity of the soil was assumed to be 0.05% by volume, half of which was continuous and the rest discontinuous. Two rainfall sequences with two initial soil water contents, evaporation versus transpiration, macropore radius ranging from 1.0 to 0.125 mm, and three different chemicals were evaluated. Over a five-week period, weekly rainfall of 25.4 mm in one hour, with soil water redistribution and evaporation or transpiration occurring between storms, generated no macropore flow when the soil was initially dry (–1500 kPa). A slight amount of macropore flow was generated under the same rainfall when the soil was initially wet (–33 kPa). Doubling the weekly rainfall amount and intensity generated macropore flow varying between 30 to 50% of rainfall depending on initial and boundary conditions. Chemicals transported with this flow were 0.05 to 8% of the surface-applied amount, depending on conditions and type of chemical. A moderately adsorbed chemical (Atrazine) was the most susceptible to macropore transport, followed in order by a strongly adsorbed chemical (Prometryn), and a mobile chemical (Nitrate). The flow entering the macropores was partially absorbed by soil at progressively deeper depths; it increased the water content of the root zone, and created a tail of low concentrations in the soil chemical content distributions. The macropore size had very little effect on macropore flow and transport, but the smallest size pores retarded the downward chemical movement by wall adsorption a little more than the largest size pores. Surface evaporation decreased macropore flow, soil water contents, and downward chemical movement, but increased chemical content of the macropore flow. Transpiration, on the other hand, decreased both macropore flow and its chemical content. Thus, this modeling study gives very useful insights into the macropore flow behavior that are very difficult to obtain experimentally, and which will be useful in characterizing macropore flow in the field.
Keywords. *Macropore, Flow.*

The term "macropores" is used here to refer to relatively large noncapillary pores or channels, such as worm holes, decayed root channels, interpedal voids, and drying cracks in clay soils. Most natural soils contain some macropores. Recent reviews of the experimental work have indicated that water flow and transport in soils containing a network of macropores can deviate substantially from predictions of the current theories (for these processes), which hold adequately in soils without a large number of macropores (Thomas and Phillips, 1979; Beven and Germann, 1982; White, 1985; Singh and Kanwar, 1991). Macropores allow rapid gravitational flow of the free water available at the soil surface or above an impeding soil horizon, thus bypassing the soil matrix. Short circuiting to groundwater through macropores is of serious concern at present, because of the possibilities of rapid transport of a portion of pesticides and other chemicals applied on the soil surface. This concern has been exacerbated by the growing practice of minimum or no tillage, for two reasons: (1) this practice entails greater pesticide use and both pesticides and fertilizer chemicals are applied on the soil surface with minimum incorporation into the soil, thus increasing soluble chemical amounts in surface flow that can enter macropores; and (2) plant residues on the surface and no tillage enhance worm activity and allow worm holes and other macropore channels to stay open at the surface. Conventional tillage, consisting of plow-disc-harrow, under most conditions destroys macropore continuity at the surface, thus considerably reducing entry of overland-flow solution into pores (Bouma et al., 1982).

Physically, the amounts of water and chemical entering macropore channels at the surface depend upon the amount of available overland flow (rainfall excess over infiltration), the amount of chemical that is transferred from surface soil to overland flow by mixing and raindrop impact, and the flow capacity of the macropores (dependent upon their number, sizes, and continuity). As a result, transport through macropores will depend upon the type of soil, biological activity, and surface conditions (especially the degree of tillage), rainfall intensities and amounts, crop water uptake, surface evaporation, and the type of chemical. Watson and Luxmoore (1986) found that under ponded flow, 73% of the flux was conducted through macropores (i.e., pores greater than 1.0 mm in diameter). Because of our concerns for groundwater

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quality, it is critical for us to know more accurately the importance of macropore transport under different conditions. Understanding the magnitude and characteristics of macropore transport as influenced by different factors will guide us in selecting suitable management practices to minimize adverse impacts on groundwater.

The USDA-Agricultural Research Service recently developed a Root Zone Water Quality Model (RZWQM), which includes a submodel of macropore flow and transport, along with submodels of other physical processes, as well as those of pesticide reactions and degradations, nutrient transformations, plant growth, and management-practice effects (DeCoursey and Rojas, 1990; RZWQM Team, 1992). The objective of this study is to use the RZWQM model to study magnitudes and characteristics of macropore flow and transport as influenced by important factors in a typical or an average (rather than an extreme) soil type. The factors studied were several macropore sizes and flow capacities per unit area, two initial soil moisture conditions, two sequences of weekly rainfall amounts and intensities, transpiration versus surface evaporation, and three types of chemicals. The characteristics of macropore flow simulated here are very difficult to study experimentally, but will be very useful in characterizing macropore flow in the field using properly calibrated RZWQM or other models.

RZWQM Components Dealing with Water and Chemical Transport in Soil Matrix and Macropores

The model components relevant to water flow and chemical transport through the soil matrix and macropores are summarized below. Water flow in our model is based on a simple two-domain or a bi-continuum approach. The two domains of flow are the soil matrix and macropore channels. The two domains interact through walls of the macropore channels which act as a common boundary for a source/sink term. Thus, in concept, our model is similar to the transient water flow models of Hoogmoed and Bouma (1980), Beven and Germann (1981), and Hetrick et al. (1982). The methods used are different however. The only source of water and chemicals transported through the macropores is overland flow (rainfall excess) generated at the soil surface and the chemicals it picks up from the surface soil by mixing and raindrop impact. The solution flow in macropore channels is assumed very rapid and unaffected by pore tortuosity. The solution is, however, subject to lateral absorption into the drier soil matrix. The reactive chemicals in solution are also subject to adsorption to or desorption from the macropore walls. For purposes of chemical transport, the soil matrix is further subdivided into micropore (immobile) and mesopore (mobile) zones. The micropores are very small pores inside soil aggregates or peds, which are assumed to take no part in flow during infiltration.

Vertical Infiltration of Water into the Soil Matrix

The soil profile can be homogeneous or layered. The Green-Ampt equation is used to calculate infiltration rates into the profile (Green and Ampt, 1911; Childs and Bybordi, 1969; Hachum and Alfaro, 1980):

$$V = \overline{K_s}\left(H_c + H_o + Z_{wf}\right)/Z_{wf} \tag{1}$$

where V is the infiltration rate at any given time, $\overline{K_s}$ is the effective saturated hydraulic conductivity of the wetting zone, H_c is the capillary drive or suction at the wetting front, H_0 is the depth of surface ponding, if any, and Z_{wf} is the depth of the wetting front. In early stages of rainfall, the infiltration rate calculated by equation 1 may be greater than the rainfall rate, thus it is set equal to the rainfall rate (Mein and Larson, 1973).

For a homogeneous soil or surface of a layered soil, the saturated hydraulic conductivity, $\overline{K_s}$, in equation 1 is set equal to the field-saturated hydraulic conductivity of the soil or horizon. For a layered soil profile with hydraulic conductivity decreasing with depth, the hydraulic conductivity is set equal to the harmonic mean saturated hydraulic conductivity of the wetted zone (Childs and Bybordi, 1969; Hachum and Alfaro, 1980). Thus, for such a layered soil the saturated hydraulic conductivity changes as the wetted depth increases with time. If hydraulic conductivity of a subsoil layer, in a layered soil, is greater than that of the harmonic mean of the layers above, the latter continues to govern the flow in this subsoil layer (Hanks and Bowers, 1962).

The capillary drive, H_c, in equation 1 varies from horizon to horizon, corresponding to the location of the wetting front. It is calculated from the unsaturated hydraulic conductivity-suction function, $K(\tau)$, of the wetting horizon as (Ahuja, 1983; Swartzendruber, 1987):

$$H_c = \frac{\int_0^{\tau_n} K(\tau)\, d\tau}{K_s} \tag{2}$$

where τ_n is the suction corresponding to average initial soil water content of the soil horizon.

For the purpose of calculations, the soil profile is divided into 1-cm depth increments down to the bottom of the profile. A simple and efficient scheme is used to integrate equation 1 for obtaining cumulative infiltration and time under these conditions. It involves the following steps: (i) an average infiltration rate, V, is calculated by equation 1 using Z_{wf}, corresponding to the middle of the wetting increment; (ii) the moisture deficit in this increment is calculated as the difference between the known field-saturated value for the layer (taken as 0.9 of porosity) and the initial moisture (the initial moisture of the wetting increment can change from the original value by lateral absorption from macropores below the wetting front); (iii) the time step in wetting this increment of soil is defined as the deficit divided by the average infiltration rate; and (iv) these incremental deficits and time steps are summed, thus calculating the cumulative infiltration and time, respectively. The excess rainfall or overland flow is calculated as the difference between the rainfall and infiltration in each time step.

Transport of a Chemical Within the Soil Matrix During Infiltration

In the spirit of the established phenomenon of the miscible displacement, we use a sequential partial-displacement and mixing approach (in 1-cm increments) to calculate chemical transport within the soil matrix during infiltration. Also, we have a provision for the presence of mobile and immobile flow regions, which introduces another form of preferential chemical transport within the soil matrix. The soil matrix porosity is divided into meso- and micropores based on either their input values or partitioning the soil water retention curve at a prescribed suction, such as 200 kPa. Initially and during first wetting of a 1-cm depth increment, the water and chemical in meso- and micropores are in equilibrium. During successive infiltration steps, the miscible displacement of solution in the saturated soil layers occurs only in the mesopores (mobile region). An alternative option allows for diffusion between the micro- and mesopore regions. This approach is similar to the layer model of Addiscott (1977), but the method of displacement is different.

For each infiltration step, the soil solution is displaced sequentially across the 1-cm soil increments in the manner of piston displacement. However, the volume of flow during an infiltration step is always less than the mesopore soil water content of a 1-cm increment (usually less than half). Thus, the displacement of solution in this increment is only partial. Mixing is allowed to occur within all mesopores of an increment after each displacement step. Thus, this two-stage process simulates miscible displacement in the mesopores. For a soil-adsorbed chemical, such as some pesticides or NH_4, a linear isotherm and an instantaneous equilibrium are assumed to occur between the soil solution and adsorbed phases in both meso- and micropore regions. At the end of an infiltration event, when the water movement slows down, the meso- and micropore regions are allowed to equilibrate. The kinetic option in the model allows a portion of a reactive chemical in the soil matrix to be subject to a first-order kinetic adsorption-desorption, rather than an instantaneous equilibrium between solid and solution phases.

Transfer of Chemicals to Overland Flow During Rainfall

After soil-surface saturation, rainfall extracts chemicals from the soil surface by raindrop impact. The extracted chemicals are transferred to overland flow. The extraction occurs from depths as great as 2 cm, but the contribution decreases exponentially with depth (Ahuja, 1986). This process can be modeled as accelerated diffusion (Ahuja, 1990), but we use a simpler approximation. We model the extraction process with a non-uniform mixing model (Heathman et al., 1986; Ahuja, 1986). The degree of mixing between rainwater and soil solution in the mesopores is assumed to be complete (equal to unity) at the soil surface (z = 0) and to decrease with depth as:

$$B = e^{-bz} \qquad (3)$$

where b is a parameter that depends somewhat on the soil type, surface roughness, and cover conditions. The B represents the fraction of rainwater that mixes with soil solution at any depth. The value of b was found to be close to 1.0 for a number of soils and conditions (Ahuja, 1986) when z is in cm. Equation 3 is used to compute chemical transfer to overland flow (that is available to move into the macropores or run off the site) from mesopores in the top two 1-cm soil increments. An integrated B value is derived for each 1-cm interval. During each infiltration step, after the overland flow begins, the chemical is first transferred to the rainfall increment of the time interval, part of which then infiltrates into the soil. Thus, the water that infiltrates during the time interval has the same chemical concentration as the overland flow.

Transport of Overland Water and Chemical Solution Through Macropores

The top soil horizons are assumed to have cylindrical macropore channels, and the bottom horizons planar cracks. Continuous macropores are idealized to be vertical and well dispersed within the soil matrix continuum. The continuity extends to groundwater table or to any specified depth. However, a certain number of deadend macropores are assumed to branch horizontally off the continuous pores in each soil horizon. Average volume fraction of macroporosity and average size of voids (radius of cylindrical macropores, and width of cracks) are assumed to be known. From this information, the number of pores or total length of cracks per unit area of soil is calculated. Maximum flow-rate capacity (K_{mac}) of these macropores is then calculated using Poiseuille's law assuming gravity flow (unit hydraulic head gradient). For cylindrical holes:

$$K_{mac} = N\rho g \pi r_p^4 / 8\eta \quad \text{or}$$

$$K_{mac} = p_{mac}\rho g r_p^2 / 8\eta \quad (ms^{-1}) \qquad (4)$$

For planar cracks:

$$K_{mac} = L\rho g d^3 / 12\eta \quad \text{or}$$

$$K_{mac} = p_{mac}\rho g d^2 / 12\eta \quad (ms^{-1}) \qquad (5)$$

where

ρ = density of water
g = gravitational constant
r_p = radius of cylindrical holes
d = width of planar cracks
η = dynamic viscosity of water
N = number of pores per unit area
L = total length of cracks per unit area
p_{mac} = macroporosity as a fraction of soil volume

After ponding, water and solutes available at the soil surface are allowed to flow into macropores to the limit of macropore flow capacity. For each time step, the flow is routed downward through the continuous macropores sequentially in 1-cm depth increments. In each depth increment, the macropore flow is allowed to flow into the deadend macropores, and be absorbed by the soil matrix by radial or lateral infiltration from both continuous and dead end parts of the macropores. The deadend macropores can also store the solution. The time dependent lateral absorption of the solution occurs only in the soil matrix below the transient wetting front generated by the continuing vertical infiltration into the matrix. This absorption is computed by radial or lateral Green-Ampt type equations. Maximum absorption is achieved when the soil water content in the depth increment reaches 90% of porosity. The above routing continues until the available solution within a given time step is exhausted or the lowest depth of interest is reached. Below the lowest depth, the solution is allowed to drain away freely. Record of the cumulative absorption and changing soil water content is maintained for each depth increment.

The transient radial infiltration rate from a cylindrical macropore is calculated based on the Green and Ampt approach as:

$$V_r = 2\pi K_s H_c / \ln\left(r_{wf}/r_p\right) \qquad (6)$$

where K_s and H_c are the saturated hydraulic conductivity and capillary drive terms for the soil matrix in the depth increment in question, r_{wf} is the wetted radius at any given time, and r_p is the macropore radius. The wetted radius, r_{wf}, is calculated from the quantity of water that has infiltrated. Equation 6 is not applicable for the very first time step when $r_{wf} = r_p$. V_r is then calculated by linearized Green and Ampt lateral absorption as:

$$V_r = 2\pi r_p \left[2K_s H_c \left(\theta_s - \theta_i\right) / \left(\Delta t_1/2\right)^{1/2}\right] \qquad (7)$$

where $(\theta_s - \theta_i)$ is the initial soil moisture deficit in the depth increment, and Δt_1 is the first time step.

For planar cracks, the equation used for lateral infiltration rate per unit length of crack is:

$$V_l = \left[2H_c K_s \left(\theta_s - \theta_i\right)/t\right]^{1/2} \qquad (8)$$

where all terms are as previously defined, and t is cumulative time for lateral flow. Both of the above equations are constrained by an upper limit of lateral flow, which depends upon the initial deficit and average distance between holes or cracks. For holes, the average soil volume per hole is assumed to be an equivalent cylinder, whose radius is the maximum radial distance of flow. The distance between cracks was derived as (1-Ld)/L.

Chemicals in macropore flow are absorbed with water by the soil matrix in each depth increment. However, before water and solutes flow into the deadend pores or are absorbed into the soil matrix, the reactive chemicals in the macropore flow equilibrate with the wall of the pore; resulting in either a net adsorption to the wall or desorption from the wall. The wall is assumed to consist of only 0.1 mm of soil. After absorption into the soil matrix, the chemicals are uniformly mixed and equilibrated with the soil and water in the mesopores within the depth increment.

Redistribution of Water and Chemicals After Infiltration

Between rainfall events, soil water is redistributed by a mass-conservative, finite-difference, numerical solution of the Richards' equation (Celia et al., 1987, 1990). The known or computed potential evaporation at the soil surface is used as the upper-boundary flux condition until the soil surface reaches a suction of 1500 kPa; thereafter, this suction is maintained at the surface as a constant head. Root water uptake from different layers is included in the equation as a sink term. Root water uptake rate is a function of potential transpiration demand of the atmosphere, plant canopy cover, root distributions and density, and soil hydraulic properties (Ahuja and Nielsen, 1990). In contrast with infiltration where computational depth increments are held uniform at 1 cm, the size of computational depth increments during redistribution increases with depth from 1 cm to 10 cm (done to reduce numerical computations). The chemicals in solution move with water from one depth increment to the other, including upward movement due to evaporation. At the end of each time step, chemical concentrations in solution and solid phases are equilibrated within each numerical depth increment. As noted earlier, the chemicals in the meso- and micropores are assumed to be in equilibrium during the redistribution. Hence, these pore classes are combined and not separately treated.

Pesticide Degradation and Other Losses

The model incorporates dissipation of pesticides from plant and crop residue surfaces, from soil surface, and from within the root zone. In each case, state-of-the-art representations of these processes are provided. For example, dissipation from soil is broken down into individual pathways, such as microbial degradation, volatilization, photolysis, hydrolysis, oxidation, and complexation. If data are not available to use individual pathways, dissipation is treated as lumped. Lumped dissipation can be treated as a simple exponential (pseudo first-order) decay function or as a double exponential. The temperature effects on dissipation are accounted for. The soil moisture effects are also approximately accounted for where important. Plant or microbial uptake of chemicals are allowed, but not considered in this study.

Heat Transport and Soil Temperature

The heat transport submodel has two components. During a rainfall event, the convective heat transported with infiltrating water is simulated like chemical transport, but the heat capacities of water and soil particles are taken into account. The temperature of rain water is assumed equal to minimum air temperature on the day of rainfall. Between rainfall events, the heat transport is simulated by a numerical solution of the convective-diffusion heat equation, using water fluxes obtained from numerical solution of the Richards' equation. The soil surface boundary condition was taken to be a constant temperature, equal to the day's mean air temperature. The lower boundary condition at 2-m depth was also a constant temperature, equal to a seasonal average value at that depth.

Soil Conditions and Other Factors Simulated

The soil selected for this study was a silty clay loam of Kirkland series (fine, mixed, thermic, Udertic Paleustoll), located at El Reno, Oklahoma. The soil profile was assumed homogeneous for this study. The soil water retention and related physical properties were measured on undisturbed soil cores at several sites (Ahuja et al., 1988). Field-saturated hydraulic conductivity was determined *in situ* (Naney et al., 1988). The mean soil-water content-suction curve, $\theta(\tau)$, of the selected soil layer was represented by a Brooks and Corey (1964) type function:

$$\begin{aligned} \theta(\tau) &= 0.473\,; & \tau \leq 12.0\ (\text{cm}) \\ &= 0.473\,(\tau/12.0)^{-0.113}\,; & \tau > 12.0\ (\text{cm}) \end{aligned} \qquad (9)$$

in which the suction, τ, is expressed in cm of water.

The $K(\tau)$ relationship, obtained from the above $\theta(\tau)$ and the field-measured K_S, was similarly expressed as:

$$\begin{aligned} K(\tau)\,(\text{cm h}^{-1}) &= 1.33; & \tau \leq 12.0\ (\text{cm}) \\ &= 1.33\,(\tau/12.0)^{-2.39}; & \tau > 12.0\ (\text{cm}) \end{aligned} \qquad (10)$$

The soil was assigned a macroporosity of 0.05% by volume. This value of macroporosity was within the range reported in the literature (Watson and Luxmoore, 1986), and the flow rate through it was large enough to allow no runoff at the highest rainfall intensity used in this study (5.0 cm h^{-1}). Half of the macroporosity consists of continuous macropores, and the other half deadend pores. With the total macroporosity fixed at this value, four different macropore sizes were simulated. The topsoil was assumed to have cylindrical macropores, and the subsoil, planar cracks. The radius of the cylindrical macropores in the top 15 cm of soil and width of cracks in the subsoil from 15 cm to 150 cm (assumed bottom of the profile) was 1.0, 0.5, 0.25, or 0.125 mm. These pore sizes are commonly thought to be macropores (Watson and Luxmoore, 1986). A simulation was also made for the soil without macropores. Microporosity was assumed to be very low, 10^{-5} m^3m^{-3}.

Two initial soil water contents in the 150 cm profile, corresponding to suctions of 33 and 1500 kPa (a moderately wet and a very dry condition respectively), were simulated for each macropore-size level. Also, in each case the soil profile was either subject to a potential evaporation of 0.48 cm d^{-1} or potential transpiration of the same magnitude. A fixed but exponentially decreasing root density distribution with depth was assumed. Selected simulations were made with both evaporation and transpiration set to zero.

Two different sequences of rainfall amounts and intensities were applied. The first sequence, hereafter called RAIN1, consisted of a weekly rainfall event of 2.54 cm in 1 h (at 2.54 cm h^{-1}), for 5 weeks. Between storms the soil was subject to water redistribution and evaporation or transpiration. The second rainfall sequence, called RAIN2, similarly consisted of five weekly rainfall amounts, and intensities that were twice the values in the first sequence.

Before the first rainfall event, three different chemicals were simulated as having been applied at the soil surface, at the rate of 4 kg ha^{-1} of active ingredients, and mixed with top 1 cm of soil. Characteristics of these chemicals are given in table 1. The chemicals varied from one that is not adsorbed to one that is strongly adsorbed by the soil; half lives were very high to moderate. Degradation rate constants were taken from the literature. The adsorbed chemicals were assumed to be in instantaneous equilibrium with soil solution; no kinetic release was simulated. To calculate soil partition coefficients for these chemicals, the soil organic matter was taken equal to 4% by weight. Since nitrate was being used as a tracer for a very mobile non-adsorbed chemical, plant uptake was not simulated.

Table 1. Characteristics of chemicals studied

Name	Partition Coefficient for Organic Carbon (K_{oc}) (L/g)	Approximate Half Life (days)
Nitrate	0.0	10^5
Atrazine	100.0	60
Prometryn	500.0	60

In running the heat transport and soil temperature submodel, the initial soil temperatures and daily air temperatures corresponded to the conditions prevailing at El Reno, Oklahoma during the months of April and May. The heat capacity and diffusion coefficients of soil were calculated from its textural composition (De Vries, 1963).

Results and Discussion

Overall Water Flow and Chemical Transport in Macropores

The cumulative amounts of water and chemicals transported in macropores during the two rainfall sequences are summarized in table 2 for most of the conditions simulated. During the RAIN1 sequence (12.7 cm total rainfall in five weekly storm events), no macropore flow was generated when the soil was initially dry (−1500 kPa), under either the evaporation or transpiration boundary condition. Even when the soil was initially wet (−33 kPa), only a very small amount of macropore flow and chemical transport occurred under the above boundary conditions. However, under wet initial conditions with no evaporation or transpiration, about 3% of the total rainfall flowed through the macropores. This situation may be likened to an initially wet no-till soil, having a good residue cover. Even in this situation, the amount of chemical entering the macropores was rather small, 0.13% of the amount applied or less for the adsorbed chemicals Atrazine and Prometryn, and much less for nitrate.

During RAIN2 sequence (25.4 cm of rainfall in five weekly storms), macropore flow was generated under all conditions (whenever macropores were present). Under dry initial soil conditions, maximum macropore flow (42% of the total rain) occurred with no evaporation or transpiration. Under conditions of evaporation and transpiration, macropore flow was 33 and 30%, respectively, of the rainfall amounts. Under wet initial soil conditions, macropore flow was greater, in all cases, than with dry soil conditions (48 to 39%). Evaporation was more effective in reducing macropore flow than transpiration. Under both initial soil water conditions, amount of chemicals transported (on a percent basis) in macropores was much smaller than the amount of water in

Table 2. Cumulative amount of water entering the macropores, other water balance components, and amounts of chemicals transported in macropores under different conditions (Data for only the largest and smallest macropore size are presented)

Rainfall Sequence – Total Amount for Five Rainfall Events (cm)	Initial Soil Water Pressure (kPa)	Upper Boundary Condition Evaporation (E) Transpiration (T) or None (0.)*	Radius of Macropores (if present) (cm)	Flow Entering the Macropores (if applicable) (cm)	Runoff (cm)	Seepage Below 150 cm (cm)	Actual Evaporation or Transpiration (cm)	Macropore Flow as Percent of Rainfall (%)	Amount of Chemicals Transported in Macropores as a Percent of the Amount Applied at the Surface (40 μg / cm²)		
									Atrazine (%)	Prometryn (%)	Nitrate (%)
Rain 1	– 1500	0.	---	---	0.086	0.4×10^{-4}	0.	---	---	---	---
12.7		E	---	---	0.0	0.4×10^{-4}	6.04	---	---	---	---
		E	0.10	0.	0.	0.4×10^{-4}	6.04	0.	0.	0.	0.
		E	0.0125	0.	0.	0.4×10^{-4}	6.04	0.	0.	0.	0.
		T	---	---	0.	0.4×10^{-4}	9.44	---	---	---	---
		T	0.10	0.	0.	0.4×10^{-4}	9.44	0.	0.	0.	0.
		T	0.0125	0.	0.	0.4×10^{-4}	9.44	0.	0.	0.	0.
	– 33	0.	---	---	0.421	3.24	0.	---	---	---	---
		0.	0.10	0.426	0.	3.48	0.	3.35	0.13	0.125	0.57×10^{-3}
		0.	0.0125	0.426	0.	3.48	0.	3.35	0.13	0.125	0.57×10^{-3}
		E	---	---	0.006	0.50	8.87	---	---	---	---
		E	0.10	0.006	0.	0.50	8.87	0.05	0.01	0.35×10^{-2}	0.55×10^{-3}
		E	0.0125	0.006	0.	0.50	8.87	0.05	0.01	0.35×10^{-2}	0.55×10^{-3}
		T	---	---	0.006	0.22	13.82	---	---	---	---
		T	0.10	0.006	0.	0.22	13.82	0.05	0.01	0.35×10^{-2}	0.55×10^{-3}
		T	0.0125	0.006	0.	0.22	13.82	0.05	0.01	0.35×10^{-2}	0.55×10^{-3}
Rain 2	– 1500	0.	---	---	9.29	0.4×10^{-4}	0.	---	---	---	---
25.4		0.	0.0125	10.67	0.	0.4×10^{-4}	0.	42.0	3.62	3.17	0.06
		E	---	---	7.38	0.4×10^{-4}	7.67	---	---	---	---
		E	0.10	8.43	0.	0.4×10^{-4}	9.05	33.2	4.62	2.67	0.13
		E	0.0125	8.48	0.	0.4×10^{-4}	9.05	33.4	4.65	2.70	0.13
		T	---	---	6.26	0.4×10^{-4}	13.14	---	---	---	---
		T	0.10	7.71	0.	0.4×10^{-4}	13.74	30.3	2.65	2.22	0.05
		T	0.0125	7.75	0.	0.4×10^{-4}	13.74	30.5	2.67	2.25	0.05
	– 33	0.	---	---	11.03	4.81	0.	---	---	---	---
		0.	0.10	12.28	0.	13.61	0.	48.3	6.52	4.15	0.82
		0.	0.0125	12.33	0.	13.61	0.	48.5	6.57	4.17	0.82
		E	---	---	8.78	0.93	9.89	---	---	---	---
		E	0.10	9.97	0.	5.07	11.09	39.2	7.85	3.57	1.30
		E	0.0125	10.02	0.	5.07	11.09	39.4	7.90	3.60	1.32
		T	---	---	9.11	0.35	13.82	---	---	---	---
		T	0.10	10.88	0.	2.98	13.82	42.8	5.77	3.62	0.82
		T	0.0125	10.93	0.	2.98	13.82	43.0	5.82	3.65	0.82

* Potential evaporation or transpiration imposed was 0.48 cm / day.

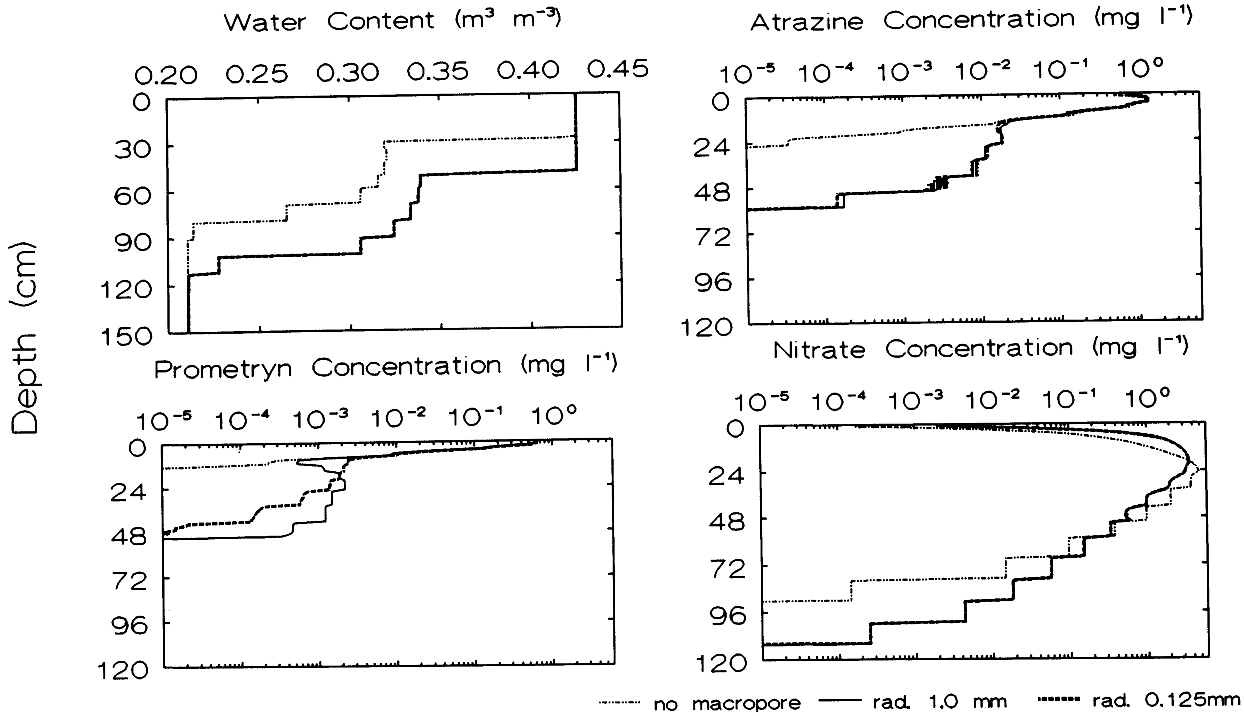

Figure 1–Water content and chemical concentration distributions in soil after application of 25.4 cm of rainfall, comparing results with no macropores vs. macropores of two extreme sizes. Initial soil water potential was –1500 kPa and the potential evaporation imposed at the soil surface was –4.8 mm d^{-1}.

all cases, 0.05 to 7.90%. Atrazine, a moderately adsorbed herbicide, was most easily transported of all chemicals, 2.65 to 4.65% of that applied under the dry initial conditions and 5.77 to 7.90% under wet conditions. Prometryn, a more strongly adsorbed chemical, had intermediate amounts transported (2.22 to 3.17% and 3.57 to 4.17%, respectively), whereas the nonadsorbed nitrate was least subject to macropore transport (0.05 to 0.13% and 0.82 to 1.32%, respectively). For Atrazine and nitrate, the macropore chemical transport was maximum under the evaporation boundary condition and least under the transpiration condition. For Prometryn, the transport was greater under the no-evaporation condition. For Atrazine this is because the material is adsorbed and thus remains on the surface; but it is not so tightly adsorbed that little is transferred into the surface water. Nitrate transfer to runoff was relatively higher under evaporation conditions because the surface evaporation tended to pull the nitrates back to the surface until it was leached below the active surface layers. The Prometryn response, because it is so highly adsorbed, is conditioned mostly by the difficulty of transfer to runoff, or macropore flow (note the values are nearly proportional to the volumes of surface runoff). For the fixed macropore volume fraction, the smaller macropore size increased water and chemical flow into macropores very slightly; the effect is practically negligible, about 0.5%.

Table 2 also shows surface runoff and the seepage of water below 150-cm depth, the bottom of the profile. No surface runoff occurred when the macropores were present under the conditions investigated. Without macropores, the runoff was higher under wetter soil and higher rainfall conditions (as expected). It was maximum when there was no evaporation or transpiration. The transpiration decreased runoff more than evaporation. For the dry initial soil condition, the seepage consisted of flow out of the soil matrix only, with or without macropores. All the macropore flow was laterally absorbed by the dry soil in the upper part of the profile, and never changed water contents near the bottom of the profile. Thus, the total seepage out of the bottom over a five-storm period was small (0.4×10^{-4} cm) and essentially the same for all conditions. However, under the wet initial soil condition, the soil profile did get wetted to the bottom, and the seepage consisted of increased flow out of the soil matrix as well as flow out of the macropores. The seepage increased with the amount of rainfall as expected. The seepage was maximum with no evaporation or transpiration. The transpiration decreased seepage more than did evaporation, both with or without the presence of macropores.

Effect of Macropore Transport on Water and Chemical Content Distributions in Soil

Water content and chemical distributions in the soil at the end of the fifth weekly rainfall event, with no macropores versus macropores of the two extreme sizes, are shown in figure 1 for RAIN2 sequence, dry initial soil and evaporation conditions. The soil profile water content is higher with macropores than without macropores because of the lateral infiltration of surface water deeper into the soil profile through macropore flow. The macropore size has essentially no effect on depth of penetration, however the chemical concentration patterns with depth are different. The Atrazine and Prometryn (adsorbed chemicals) concentration distributions have a distinct tail as a result of deeper transport of those chemicals through macropores. The concentrations in the

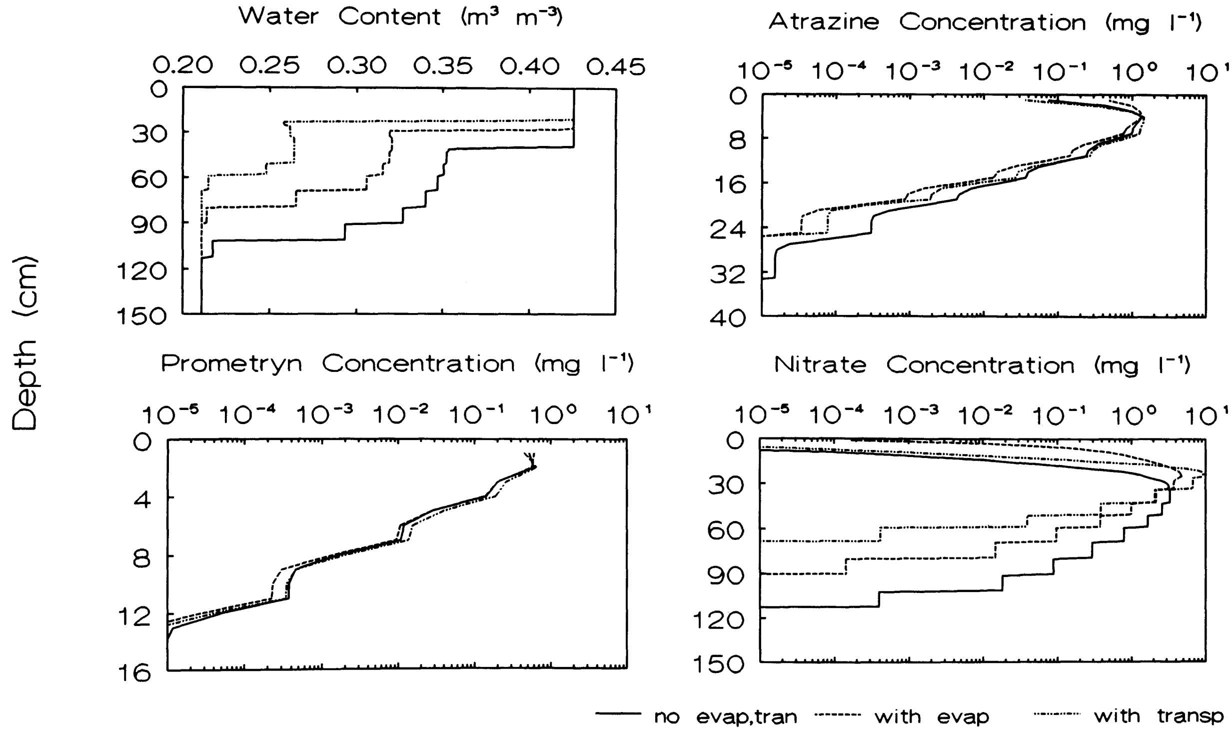

Figure 2–Comparison of soil moisture levels and chemical concentrations with no macropores and with and without evaporation and transpiration. Plots show conditions after application of 25.4 cm of rainfall. Initial soil water potential was –1500 kPa, evaporation and transpiration rates (when imposed) were –4.8 mm d^{-1}.

tail region are, however, small. (Note the log scale of the concentrations on graphs.) The smaller macropore size causes some retardation in the Prometryn movement in the tail region, although overall amount in the tail is slightly increased. The effect is less in the less strongly adsorbed Atrazine. This retardation is due to adsorption on macropore walls; the smaller size pores have greater surface area per unit of soil surface area for adsorption for a given volume fraction of macropores. The soil surface concentration of the above chemicals was also slightly higher with macropores than without macropores. For nitrate, the distribution is spread out more with macropores than without macropores, the concentrations being higher on both arms of the pulse. All of the above differences are due in part to a smaller fraction of water moving through the soil matrix under macropore soil conditions.

Under the same initial and boundary conditions as above — except transpiration instead of evaporation — the general features of the water content and chemical distributions, with and without macropores, were essentially the same as described above. However, the chemical amounts forming the tail were smaller than with evaporation. On the other hand, the wet-initial conditions under RAIN2 increased the concentrations of chemicals deeper in the soil profile.

Effect of Evaporation vs. No Evaporation. In all cases, soil water contents in the profile were lower with evaporation than without evaporation (e.g., fig. 2). Evaporation resulted in less downward movement of water, and thus retarded downward movement of all chemicals, and caused higher concentrations near the soil surface. This effect was greater for the mobile chemical, nitrate, and least for the strongly adsorbed Prometryn (see fig. 2). Until infiltration leached the nitrate and Atrazine below the zone influenced by evaporation, higher surface concentrations caused by evaporation, increased the amounts of Atrazine and nitrate entering the macropores with surface water as compared with no-evaporation, even though there were greater amounts of water entering the pores under the no-evaporation condition (see table 2).

Effect of Transpiration vs. No Transpiration. The soil water contents in the soil profiles are much lower with transpiration (root uptake of water) than without transpiration or evaporation (e.g., fig. 2); in fact, even lower than with evaporation. Thus, transpiration decreases downward movement of water and chemicals even more than evaporation. The maximum effect is on the movement of nitrate (fig. 2). This effect was present under both rainfall sequences and initial conditions. Transpiration did not increase the chemical concentrations near the soil surface, as did evaporation. As a consequence and with less downward water movement, transpiration generally decreased the chemical transported in macropores, as compared to either evaporation or nontranspiration and no-evaporation. The distribution of nitrate concentration under transpiration was less dispersed than with evaporation.

Water Flow and Chemical Transfer from Macropores to Soil Matrix at Different Depths — Effect of Macropore Size

Water flow from continuous and discontinuous macropores to the soil matrix at different depths during each of the five rainstorms of the RAIN2 sequence are presented in figure 3, for the largest and smallest macropore sizes simulated. In the case of dry initial soil and with evaporation, all of the lateral flow occurred

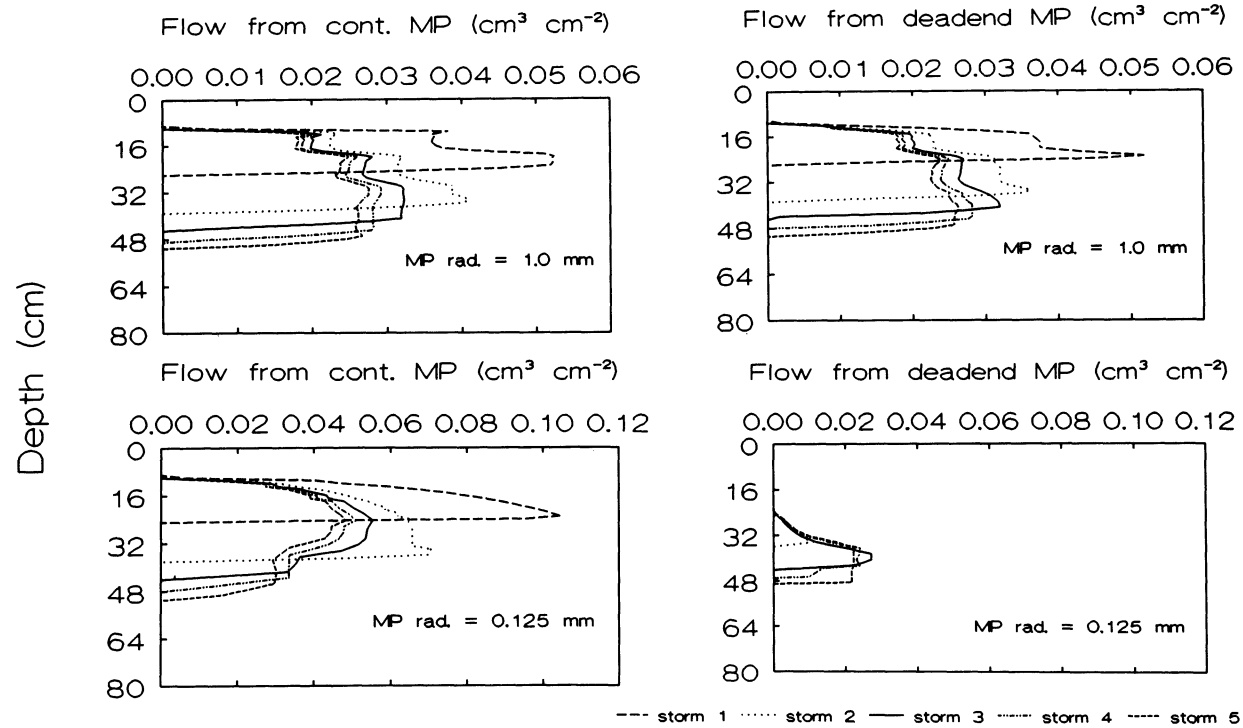

Figure 3–Water flow from continuous and deadend macropores (MP) of the two extreme macropore sizes to the soil matrix as a function of depth for each of the five 5.08 cm rainstorms. Initial soil water potential was –1500 kPa and potential evaporation imposed at the soil surface was –4.8 mm d^{-1}.

within the top 50 cm of soil. Pore water flow reached progressively deeper with each successive rainstorm as the soil water deficit was relieved, but the average magnitude of flow per centimeter depth decreased. For a macropore size of 1.0 mm, the successive flows from deadend macropores were only slightly smaller than from continuous macropores (see fig. 4). For the smaller size macropores of 0.125 mm, flow from the deadend macropores was much smaller than the flow from continuous macropores (see fig. 4); up to about 20 cm depth, all the available water was transferred to the soil matrix through continuous macropores, none through deadend pores. This difference in flow patterns between the two macropore sizes is due to the fact that the smaller size pores have overall greater surface area, and hence greater absorption capacity for water, than the larger-size pores, at a given level of macroporosity.

The Atrazine transfer from macropores to soil matrix, corresponding to the water flows shown in the above graphs, are presented in figure 5. Partitioning of transfer between the continuous and discontinuous macropores is similar to that of water flow, except that for the smaller macropore size the chemical transfer from deadend pores to soil matrix is a factor of 5 to 7 less than from continuous pores. This latter effect is apparently caused by greater overall adsorption of the chemical on the walls of continuous pores when the pores are of smaller size than when the pores are larger. Figure 6 shows that the Atrazine adsorbed on macropore walls per unit cross section of soil was an order of magnitude higher in the case of smaller-size pores.

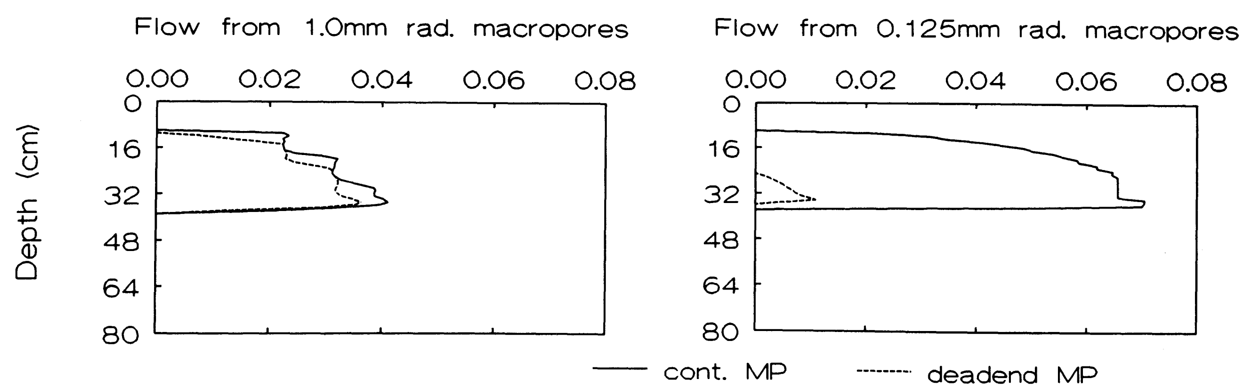

Figure 4–A comparison of flow from continuous and deadend macropores 1.0 and 0.125 mm in radius during the second application of 5.08 cm of water. Initial soil water potential was –1500 kPa and potential evaporation imposed at the soil surface was –4.8 mm d^{-1}.

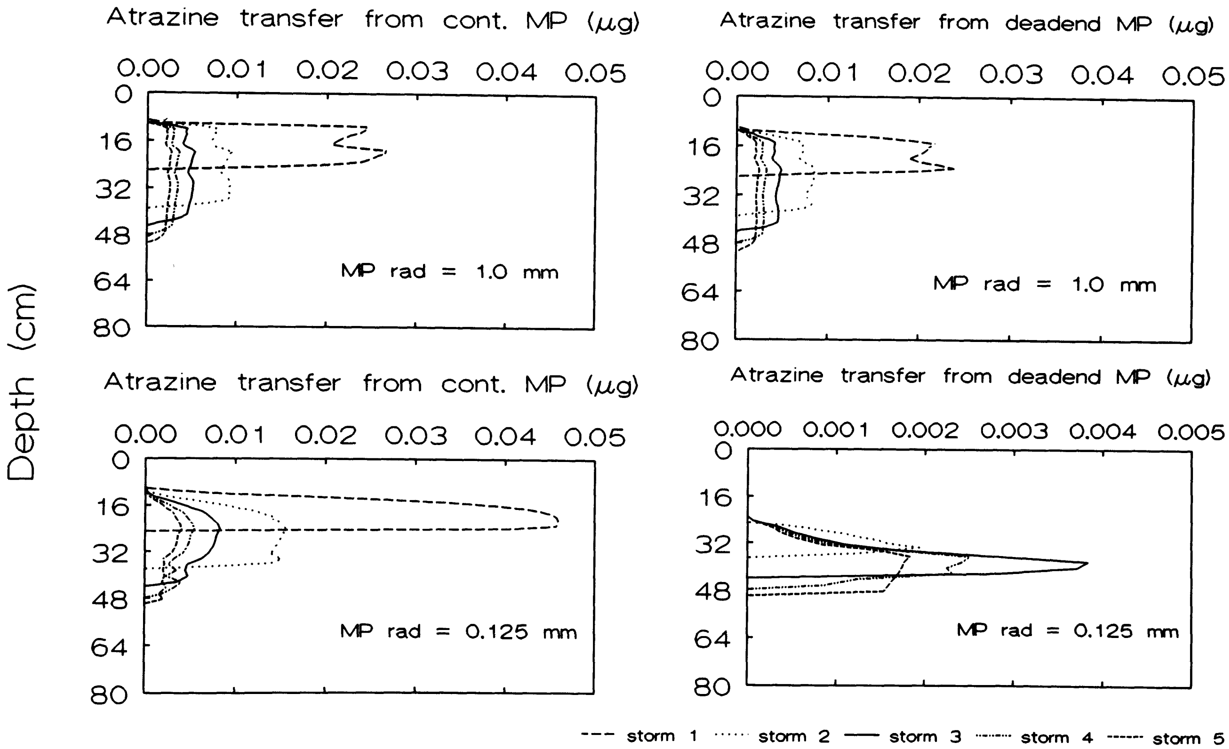

Figure 5–Atrazine transfer from macropores to soil matrix at different depths during each of five 5.08 cm rainstorms, for the two extreme macropore sizes. Initial soil water potential was –1500 kPa and potential evaporation imposed at the soil surface was –4.8 mm d^{-1}. Note difference in scale of atrazine concentrations for macropores of size 0.125 mm.

The Prometryn transfer from macropores to soil matrix is shown in figure 7. The profiles are similar to those of Atrazine, except that the amounts of transfer are smaller. For macropore size of 1.0 mm, the amounts are about half those of Atrazine. But for macropore size of 0.125 mm, the Prometryn amounts transferred are more than an order of magnitude smaller than corresponding Atrazine amounts. These differences are caused by smaller concentrations of Prometryn in water entering the macropores with overland flow, and much more retention on macropore walls in case of the smaller pore size; both of these effects being due to a much stronger soil-adsorption coefficient of Prometryn than of Atrazine. The Prometryn amount retained on macropore walls was about twice the amount of Atrazine retained on macropore walls. For the nitrate transfer, the profiles were similar and closer to those of Atrazine.

For other RAIN2 sequences, i.e., with transpiration, no-evaporation, and/or the wet initial soil conditions, the results of water flow and chemical transfer from macropores to soil matrix were similar in trends to those presented above, but the magnitudes and vertical distributions were different.

Conclusions and Inferences

Simulations with RZWQM bring out some interesting and useful results and implications. The fact that no or negligible macropore flow was generated by five weekly rainfalls of 25 mm each in a medium heavy soil, indicates

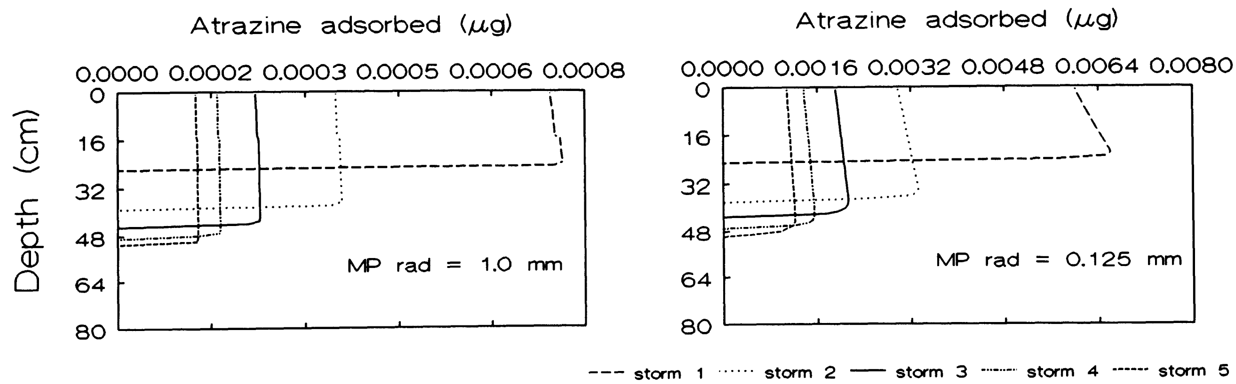

Figure 6–Atrazine adsorbed on macropore walls per cm of soil at different depths, for two macropore sizes. Initial soil water potential was –1500 kPa and potential evaporation imposed at the soil surface was –4.8 mm d^{-1}. Note difference in scale of atrazine concentrations.

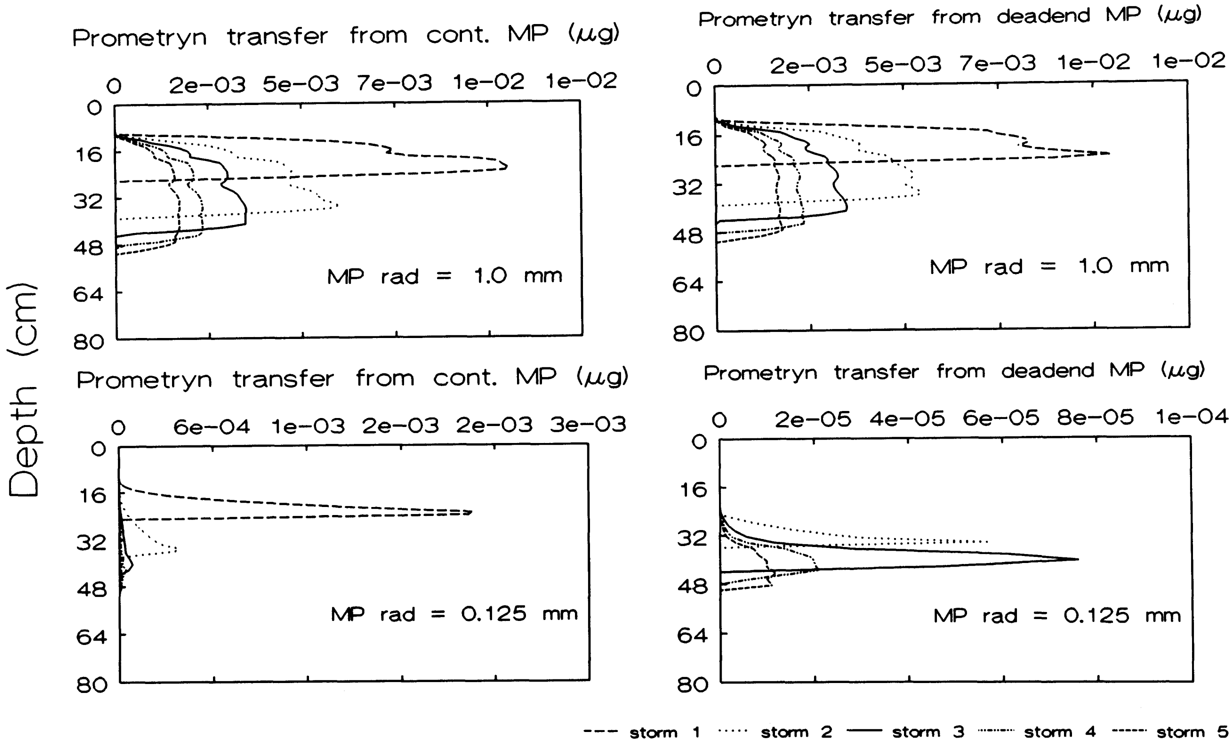

Figure 7–Prometryn transfer from macropores to soil matrix at different depths during each of five 5.08 cm rainstorms, for the two extreme macropore sizes. Initial soil water potential was –1500 kPa and potential evaporation imposed at the soil surface was –4.8 mm d^{-1}. Note difference in scale of prometryn concentrations for macropores of size 0.125 mm.

that the macropore flow and transport may not occur during many of the natural rain events. Under heavier natural rainfall amounts, a lot more water will flow through macropores (30 to 50% of the applied rainfall in our study), but the percentage of surface-applied chemicals transported with this water will be relatively much smaller (0 to 8% in our study). Moderately adsorbed herbicides (e.g., Atrazine) are most susceptible to transport through macropores, and thus should be the target of preventive management practices. Generally, nitrate amounts entering the macropores will be small, except when there is a very heavy rainfall immediately after application of the fertilizer. However, the additional water infiltrating through macropores, its lateral absorption by the soil matrix, and subsequent redistribution, do result in some nitrate moving deeper into the profile. Overall, the absorption of solution in macropores by the soil matrix and adsorption of chemical ions on macropore walls retards the quick short-circuiting of solution entering the macropores to groundwater.

The finding that the macropore size has only a small and negligible effect is very important and useful; this makes the field characterization of the macropore flow and its use in modeling a lot easier. One needs to know just the overall maximum macropore flow rate, without knowing the size distribution of macropores. A reasonable average pore size can be assumed. Finally, this modeling study clearly brings out the role of evaporation versus transpiration on the transport of chemicals through macropores and the soil matrix. Evaporation increases the amount of chemicals transported in macropores, but decreases their downward movement through the soil matrix, as compared with no evaporation. Thus, in a no-till soil containing continuous macropores, the surface residue cover may actually decrease the transport of herbicides in macropores, but increase their downward movement through the soil matrix. On the other hand, the water uptake by roots (transpiration) decreases both the amount of chemical entering the macropores and the movement in the soil matrix; a distinct advantage of cover crops.

The above conclusions and inferences are based on our study with a silty clay loam soil, but can be qualitatively extrapolated to other soil types. In lighter soils (loams or sands) of higher permeability, macropore flow and transport will be less in overall magnitudes than in our soil, whereas in a heavier clay soil they will be more. This is simply due to the fact that when the hydraulic properties of the soil matrix allow higher rates of infiltration, there is less of the rainfall excess available at the soil surface to enter the macropores. A soil surface crust formed by rainfall will increase macropore flow, unless the sediment splash during the process of crust formation plugs up the macropore openings to the surface. The various assumptions made in our model concerning macropores are, in our judgement, reasonable at this point in our knowledge. However, further experimental studies on the nature and geometry of macropores and the nature and dynamics of flow in them are certainly needed.

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